

Lab 2: Sensor Technology (ELA400)

Practical Assignment: Antenna Measurement and Characterization

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Name:

Date:

I. ABSTRACT

In this practical lab you will familiarize yourself with the basic principles of antenna measurements and how to evaluate antennas with respect to their efficiency, radiation pattern and polarization. The lab involves measurements of the input and output port **voltage reflection coefficients** (S_{11} and S_{22} , respectively), as well as the **forward voltage gain** (S_{21}) of a two-antenna setup under different conditions. All measurements will be taken with a Vector Network Analyzer (VNA) in the frequency range of **2 to 3 GHz**. Conclusions are then being drawn about the antenna characteristics based on the **transmitted and reflected voltage waves** at different frequencies. The lab involves preparatory tasks that are to be performed at home, while measurements and evaluation of the result are conducted in the research lab under the supervision of the lab instructors.

II. PREPARATORY TASKS

1. In order to familiarize yourself with common values used in antenna and microwave engineering and how these values are related to each other, complete the following table:

VSWR	S_{11}	Γ	RL
1	-inf	0	inf
1.01	-46dB	0,005	46dB
1.02	-40,1dB	0,0099	40,1dB
1.065	-30	0.0316	30
1,106	-26	0,05	26
1,22	-20	0,1	20
1.5	-14dB	0,2	14dB
1.6	-12,74dB	0,23	12,74dB
1,92	-10	0,32	10
10.5	-1,66dB	0,826	1,66dB
19	-0.915dB	0.9	0.915dB
39	-0,446dB	0.95	0,446dB
199	-0,087dB	0.99	0,087dB
inf	0	1	0

- One of the two antenna models you will use in this lab is the **VP165/24 flat panel antenna**. Familiarize yourself with the data sheet and the main characteristics of the antenna. What are the:

Input impedance: **50 ohm**

Antenna gain: **9dbi**

Maximum VSWR: **<1.5**

Frequency band: **2400 - 2485 MHz**

- Assume an open-ended transmission line (TL) with a characteristic impedance of 50Ω . This can be interpreted as a TL that is terminated with air, which is known to have a characteristic impedance of 377Ω . What is the reflection coefficient and the S_{11} -parameter?

$$\Gamma = -0.766$$

$$S_{11} = -2.32\text{dB}$$



III. ANTENNA MEASUREMENTS

The following measurements will be performed in the research lab under the supervision of the lab instructors. You will use channel 2 and 3 of the Vector Network Analyzer (VNA), i.e, the measured S-parameters will be S_{22} , S_{33} and S_{32} . Write down all measurement results in the dedicated fields provided in this document.

- Operational Bandwidth:** Connect the black WiFi antenna and the white panel antenna to port 2 and 3 of the VNA, respectively, and place them 30 cm apart (measured at the center point of the top cover). Measure the **resonance frequencies** and the **upper and lower corner frequencies** based on the criteria **$S_{11} < -10 \text{ dB}$** for the WiFi antenna and **VSWR < 1.5** for the panel antenna. To measure these values, first select the desired **S-parameters** to monitor (S_{22} for the WiFi antenna, S_{33} or the panel antenna), **autoscale** if necessary and place markers at the three frequencies of interest. Write down the values in the dedicated spaces below.

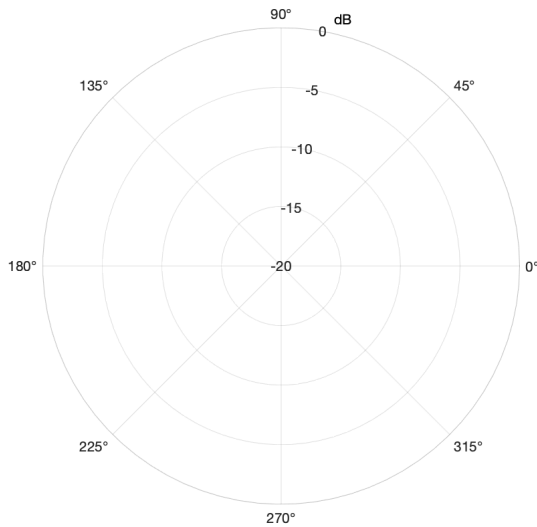
WiFi antenna:	$f_l =$	$f_c =$	$f_u =$	BW =
Patch antenna:	$f_l =$	$f_c =$	$f_u =$	BW =

Why do the bandwidths differ if taken from the S-parameters or the VSWR?

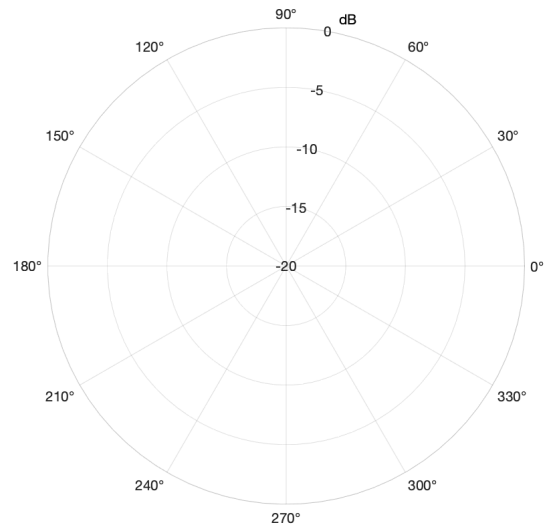
- Radiation Pattern:** Turn the WiFi antenna with the marker towards the panel antenna (corresponding to a turn of 0°) and display S_{32} in logarithmic scale on the screen. Autoscale if necessary and set a marker at the frequency of highest power transfer. This frequency will be your reference value for the next measurements. Turn the WiFi antenna clockwise in steps of 45° in the azimuth plane and note down the measured values in the table below until you performed a full turn of 360° .

Now bend the Wifi Antenna in steps of 30° towards the patch antenna (elevation plane) until you reach an angle of 90° . Write down the values in the table below.

	0°	30°	45°	60°	90°	135°	180°	225°	270°	315°	360°
Azimuth		-		-							
Elevation			-			-	-	-	-	-	-



(a) Azimuth



(b) Elevation

Insert the *normalized* values in the polar diagrams above. To normalize the values, add the maximum measured value of the data set to all data points of that set (after the normalization, the maximum value in the data set should equal 0 dB). What is the radiation pattern of the WiFi antenna? What basic antenna type does this pattern resemble?

- Power Decay:** Dismount the WiFi antenna and replace it with the second panel antenna. Let the antennas face each other and display the S_{32} parameter in logarithmic scale. Autoscale if necessary and set a marker at the frequency of highest power transmission. Write down the value in the dedicated space below. Measure the distance between the top center points of both antennas, then move the antenna at channel 2 until the distance between both antennas is doubled. Write down the S_{32} at the marker position.

	Distance (cm)	S_{32} (dB)
Measurement 1		
Measurement 2		

Assuming the S_{32} denotes the ratio of incident to transmitted **voltage**, what is the proportional relationship between electric field and distance of a spherical wave?

Assuming the S_{32} denotes the ratio of incident to transmitted **power**, what is the proportional relationship between power density and distance of a spherical wave?

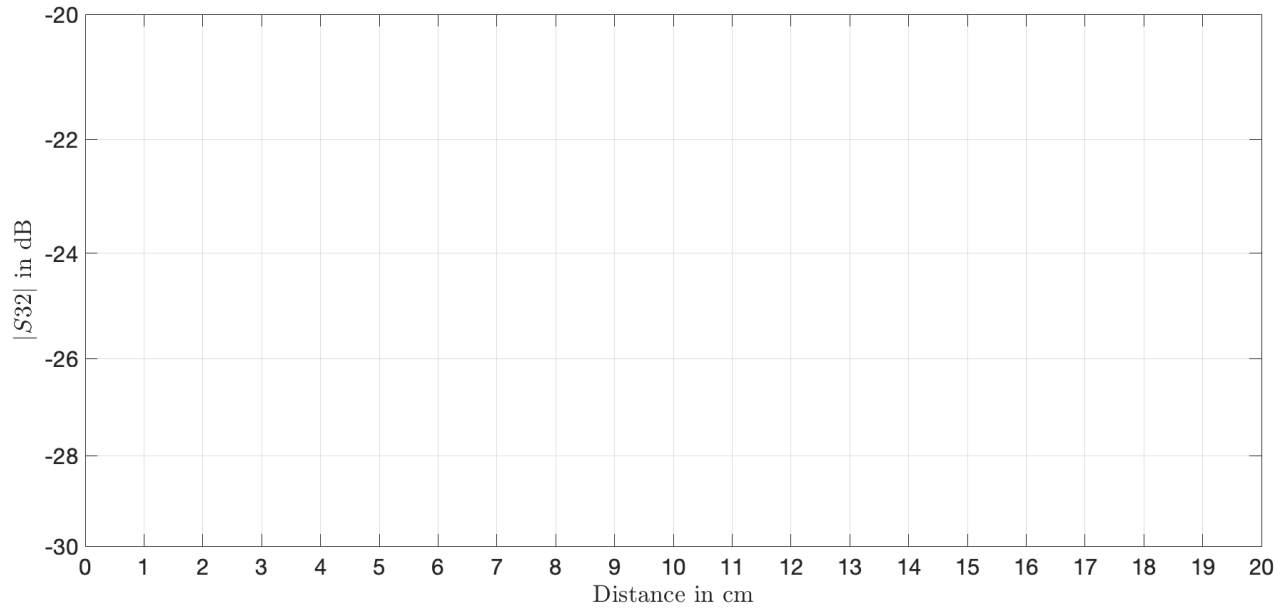
Bonus: Move the unconnected WiFi antenna between the two panel antennas, first holding the WiFi antenna vertically, then horizontally. In which orientation is the transmission more disturbed and why? Discuss how polarizers (e.g. in sunglasses) work.

- Standing Wave Pattern:** Turn the antenna connected to channel 2 by 180° and let its back face the antenna connected to channel 3. Move the channel 2 antenna as far away as possible and write down the S_{32} at 2.45 GHz. Now place a metal panel provided by the lab instructors in front of channel 2 and compare the measured values. Discuss the results.

Without metal plane:

With metal plane:

Start moving the channel 2 antenna in steps of 2 cm away from the metal sheet. After every movement, let the value stabilize for some seconds and mark the measured value at 2.45 GHz in the provided diagram. What type of pattern do you expect to measure?



5. **Bonus: Electromagnetic Applicators:** Measure the reflection coefficient of one panel antenna at 2.45 GHz. Place your hand over the radome and observe how the spectrum changes. Now connect an electromagnetic applicator provided by the lab instructor. Measure its behaviour first in air, then with the applicator touching tissue, e.g., a part of your arm or your hand. Compare the behavior of the antenna and the applicator and discuss their differences.

IV. REPORT

The preparatory tasks must be completed before the lab, while the measurements and discussion of the results will take place in the research lab under the supervision of the lab instructors. The lab instructions will serve as the lab report.

All members of a group must fill out their own lab report. At the end of the lab, all students hand in their individual report in order to pass the lab.

REFERENCE LITERATURE

- [1] D. M. Pozar, *Microwave engineering; 3rd ed.* Hoboken, NJ: Wiley, 2005.
- [2] C. A. Balanis, *Antenna Theory: Analysis and Design.* Wiley-Interscience, 2005.
- [3] R. E. Collin, *Foundations for Microwave Engineering.* New Delhi,: Wiley,, 2 nd. ed., c2009.
- [4] D. J. Griffiths, *Introduction to Electrodynamics.* Noida: Pearson, 4th ed. ed., 2015.